

A new proof of a proposition concerning the theory of quadratic forms

Translated from the French

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A new proof of a proposition concerning the theory of quadratic forms.

The proposition of which I wish to give a new and very simple proof belongs to a theory presented in detail in Gauss's *Disquisitiones arithmeticae*; I may therefore refer to that work for the terminology I shall have to use, and pass immediately to the object to be accomplished.

Given an improper substitution

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

by which the quadratic form $(a, b, c) = f$, whose determinant is supposed different from zero, is changed into itself, one can always obtain a new form belonging to the same class, and for which twice the middle coefficient is a multiple of the first coefficient.

The supposition in the statement immediately supplies, besides the equation

$$\alpha\delta - \beta\gamma = -1,$$

the following three:

$$a(\alpha^2 - 1) + 2b\alpha\gamma + c\gamma^2 = 0,$$

$$a\alpha\beta + 2b\alpha\delta + c\gamma\delta = 0,$$

$$a\beta^2 + 2b\beta\delta + c(\delta^2 - 1) = 0,$$

where the second has been simplified by substituting $\alpha\delta + 1$ for $\beta\gamma$. Taking the sum of the first two after multiplying them respectively by $-\delta$ and γ , and then the sum of the last two after multiplying by β and $-\alpha$, one obtains the new equations

$$a[\alpha(\beta\gamma - \alpha\delta) + \delta] = a(\alpha + \delta) = 0,$$

$$c[\delta(\beta\gamma - \alpha\delta) + \alpha] = c(\alpha + \delta) = 0.$$

It follows that, if a and c do not vanish simultaneously, then

$$\alpha + \delta = 0.$$

But this last equation also holds in the excepted case; for, since in that case b is different from zero, one has at the same time

$$\alpha\gamma = 0, \quad \alpha\delta = 0, \quad \beta\delta = 0,$$

equations compatible with

$$\alpha\delta - \beta\gamma = -1$$

only if one supposes simultaneously

$$\alpha = 0, \quad \delta = 0.$$

It is therefore proved that our improper substitution is always of the form

$$\begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix},$$

where

$$\alpha^2 + \beta\gamma = 1.$$

Let us pause for a moment at the particular case in which the third coefficient γ of this substitution vanishes. Our equations then give

$$\alpha = -\delta = \pm 1, \quad 2b = \pm a\beta.$$

Thus, in this case, the given form f already has the required property. It follows that, if γ is not zero, everything reduces to obtaining a form equivalent to f , and transformable into itself by an improper substitution whose third coefficient vanishes.

For this purpose let

$$\begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

be the most general proper substitution; that is, let λ, μ, ν, ρ be four integers subject to the sole condition

$$\lambda\rho - \mu\nu = 1.$$

Let φ denote the form into which f is changed when this substitution is applied to it. The inverse substitution

$$\begin{pmatrix} \rho & -\mu \\ -\nu & \lambda \end{pmatrix},$$

which is also proper, will change φ into f .

Now compose the three substitutions

$$\begin{pmatrix} \rho & -\mu \\ -\nu & \lambda \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

in the order in which we have just written them. Since the first changes φ into f , the second changes f into f , and the third changes f into φ , the composed substitution, which is moreover evidently improper, changes the form φ into itself. The question is thus reduced to making the third coefficient of our composed substitution equal to zero.

To obtain this coefficient, first compose the first two substitutions, and write only the third and fourth coefficients of the substitution that results from this first composition. These coefficients are

$$\gamma\lambda - \alpha\nu, \quad -\alpha\lambda - \beta\nu.$$

It follows that the coefficient to be obtained is

$$(\gamma\lambda - \alpha\nu)\lambda - (\alpha\lambda + \beta\nu)\nu = \gamma\lambda^2 - 2\alpha\lambda\nu - \beta\nu^2.$$

Since γ is not zero by hypothesis, we may, before setting our expression equal to zero, multiply it by γ . We then have

$$\gamma(\gamma\lambda^2 - 2\alpha\lambda\nu - \beta\nu^2) = (\gamma\lambda - \alpha\nu)^2 - (\alpha^2 + \beta\gamma)\nu^2 = (\gamma\lambda - \alpha\nu)^2 - \nu^2 = 0,$$

and consequently

$$[\gamma\lambda - (\alpha + 1)\nu][\gamma\lambda - (\alpha - 1)\nu] = 0.$$

Thus, if after reducing the ratio $(\alpha \pm 1)/\gamma$, where the ambiguous sign is arbitrary, to its simplest expression λ/ν , one derives from the integers λ, ν two integers μ, ρ satisfying

$$\lambda\rho - \mu\nu = 1,$$

then the substitution

$$\begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

applied to the form f will change that form into another equivalent form φ having the required property.

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